

Evictable Vector Commitments: Verifiable LLM Inference under KV-Cache Eviction

Anonymous Submission

Abstract. Verifiable LLM inference lets a client check that a model was executed faithfully on its input. Existing schemes commit to the entire retained activation state and therefore cannot account for *KV-cache eviction*: the permanent deletion of token key/value states performed by streaming cache compressors such as StreamingLLM, H2O, and SnapKV. We introduce *evictable vector commitments* (EvictVC), a commitment primitive whose committed positions can be permanently evicted under a public eviction policy while a tamper-evident history of every deletion is maintained. We formalize the primitive with a syntax and three game-based security notions—eviction soundness, policy-binding, and history tamper-evidence—and give a construction from KZG polynomial commitments in which an eviction is a single group operation independent of the cache size, an audit proof for an arbitrary set of positions is a single group element, and all opening proofs are maintainable in quasi-linear time via the Feist-Khovratovich technique. We prove security under the n -SDH assumption and the collision resistance of a hash function. A prototype over BLS12-381 confirms the asymptotics: eviction and commitment updates cost $\approx 135\mu\text{s}$ irrespective of cache length up to 2^{16} , verification is two pairings ($\approx 2.6\text{ ms}$), and audit proofs are 48 bytes regardless of how many positions are audited.

Keywords: Vector commitments · KZG polynomial commitments · Verifiable computation · Machine-learning inference · KV-cache compression

1 Introduction

Delegated large-language-model inference increasingly runs on servers a client does not control. *Verifiable inference* addresses the resulting trust gap by having the server commit to its computation and prove that the claimed output follows from a faithful execution. A common ingredient is a commitment to the model’s intermediate state, against which the server later opens individual values to answer audit challenges.

A central object of that state is the *key/value (KV) cache*: the stored attention keys and values for every past token. The cache grows linearly with context length, so production systems compress it. Streaming compressors such as StreamingLLM, H2O, and SnapKV do so by *permanently evicting* the key/value states of tokens deemed unimportant under an attention-based policy. Eviction is destructive: the evicted state is gone and can never be opened again.

This destructiveness breaks existing verifiable-inference commitments, which assume the committed state remains fully present and openable. Three problems arise simultaneously. First, the commitment must be able to *shrink*: an evicted position can no longer be claimed to hold a value. Second, eviction must be *policy-bound*: a malicious server must not be able to drop an inconvenient (e.g. high-attention) position while claiming it followed the compressor’s policy. Third, the server must not be able to *rewrite history*: the record of which positions were evicted, and with what values, must be immutable so that an auditor can reconstruct and check the entire compression trajectory.

We resolve all three with *evictable vector commitments*. An EvictVC commits jointly to a value vector and a per-position score (attention) vector. It supports the usual openings

and homomorphic updates, and adds an **Evict** operation that (i) removes a position from the live value commitment in a single group operation, (ii) is accepted only if the position’s committed score satisfies the public eviction policy, and (iii) appends the removed record to a tamper-evident hash-chain history. Audits remain cheap: a proof for an arbitrary set of live positions is a single group element, and all opening proofs are maintainable in quasi-linear total time.

Contributions.

- **A model.** We give a syntax for evictable vector commitments and three game-based security notions—*eviction soundness*, *policy-binding*, and *history tamper-evidence* (Sections 4–5).
- **A construction.** We instantiate the primitive from KZG polynomial commitments over roots of unity (Π_{KZG} , Sections 6–7): eviction and updates are $O(1)$, audit proofs for any position set are a single group element, and all opening proofs are produced in $O(n \log n)$ via the Feist–Khovratovich technique.
- **Security proofs.** We give tight reductions: eviction soundness and policy-binding reduce to position binding of the KZG vector commitment, hence to n -SDH; history tamper-evidence reduces to collision resistance (Section 8).
- **A prototype.** A Python reference and a Rust/arkworks backend over BLS12-381 validate the design: eviction costs $\approx 135 \mu\text{s}$ independent of cache size up to 2^{16} , audit proofs are 48 bytes versus linearly growing Merkle multiproofs, and Feist–Khovratovich makes whole-cache proof generation practical (Section 9).

Organization. Section 2 surveys related work. Sections 3–5 develop the model. Sections 6–7 give the construction and Section 8 its security proofs. Section 9 reports the evaluation and Section 10 concludes.

2 Related Work

Verifiable and accountable LLM inference. The strongest guarantee is a zero-knowledge proof of the forward pass: zkLLM [SLZ24a] is the first zero-knowledge argument tailored to LLMs, proving an entire 13-billion-parameter inference in about fifteen minutes with a proof under 200 kB while keeping the model parameters hidden, via a custom lookup argument for the non-arithmetic tensor operations. Such proofs are sound but impose orders-of-magnitude prover overhead. A lighter line trades cryptographic soundness for empirical or economic guarantees: VeriLLM [W⁺25] commits to sampled activations with Merkle trees and re-executes roughly 1% of the computation under a one-honest-verifier assumption, deterring lazy verification with a peer-prediction incentive, while SVIP [SLZ⁺24b] and TopLoc [OFP⁺25] detect deviation through lightweight activation fingerprints and locality-sensitive hashing. A feature common to all of these is that the prover commits to the *retained* computation—activations or transcripts that remain available to be opened or re-run. None models the *permanent deletion* of committed state, which is precisely what streaming cache compression performs and what EvictVC targets. Moreover, the Merkle-tree commitments several of these systems use yield audit proofs that grow with the number of checked positions, whereas EvictVC’s subvector audits are constant-size and aggregatable (Section 9).

Vector commitments. Vector commitments with position binding were formalized by Catalano and Fiore [CF13]; subsequent work added single-element subvector openings

and cross-commitment aggregation [TAB⁺20] as well as efficient local updates. EvictVC extends this lineage with a deletion-style, policy-checked eviction operation and an append-only, tamper-evident history, while reusing single-element subvector openings to make batch audits constant-size.

KV-cache compression. Streaming cache compressors permanently evict the key/value states of low-importance tokens to bound memory. StreamingLLM [XTC⁺24] keeps “attention-sink” and recent tokens under a positional window; H2O [ZSZ⁺23] scores tokens by cumulative attention and retains the “heavy hitters”; SnapKV [LHY⁺24] selects clustered important tokens from a prefill observation window. In each case the unselected token states are removed from memory and become permanently inaccessible—the destructive operation our primitive must accommodate—and the threshold/score structure of these policies is exactly the policy predicate that EvictVC binds. These works optimize output quality and memory; they provide no verifiability, the orthogonal axis we add.

Amortized KZG proofs. The Feist–Khovratovich technique [FK23] computes all evaluation proofs of a KZG commitment in quasi-linear total time rather than recomputing each in isolation. We apply it (Section 9) so that opening proofs for the entire, possibly-evicted cache can be produced and maintained efficiently.

3 Preliminaries and Building Blocks

Notation. We write λ for the security parameter and 1^λ for its unary encoding. A function is *negligible*, written $\text{negl}(\lambda)$, if it vanishes faster than every inverse polynomial. PPT abbreviates probabilistic polynomial time. For $n \in \mathbb{N}$ we write $[n] = \{1, \dots, n\}$. We write $x \xleftarrow{\$} X$ for sampling x uniformly from a finite set X . All algorithms implicitly take pp as input once it is fixed. Concatenation of bit strings is denoted $\|$, and $\langle \cdot \rangle$ denotes an unambiguous (prefix-free) encoding into $\{0, 1\}^*$.

Bilinear groups. For the concrete instantiation we use a Type-III bilinear group $\text{bgp} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, g_1, g_2, p)$ of prime order p , where $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a non-degenerate efficiently computable pairing. For $x \in \mathbb{F}_p$ we write $[x]_1 := x \cdot g_1$, $[x]_2 := x \cdot g_2$, $[x]_T := x \cdot e(g_1, g_2)$, so that $e([a]_1, [b]_2) = [ab]_T$. The abstract syntax of Section 4 is group-agnostic.

3.1 Vector commitments with updates

Definition 1 (Vector commitment). A *vector commitment* over message space $\mathcal{V}$ is a tuple of PPT algorithms $\text{VC} = (\text{VC.Setup}, \text{VC.Commit}, \text{VC.Open}, \text{VC.Verify}, \text{VC.Update})$:

- $\text{VC.Setup}(1^\lambda, 1^n) \rightarrow \text{pp}$ outputs public parameters for vectors of length n .
- $\text{VC.Commit}(\text{pp}, \mathbf{m}) \rightarrow (C, \text{st})$ commits to $\mathbf{m} \in \mathcal{V}^n$, returning a commitment C and committer state st .
- $\text{VC.Open}(\text{pp}, \text{st}, S) \rightarrow \pi$ produces a proof for the entries indexed by $S \subseteq [n]$.
- $\text{VC.Verify}(\text{pp}, C, S, \{m_i\}_{i \in S}, \pi) \rightarrow \{0, 1\}$ verifies the claimed entries against C .
- $\text{VC.Update}(\text{pp}, C, i, \delta) \rightarrow C'$ deterministically updates C to the commitment of the vector with m_i replaced by $m_i + \delta$.

We write $\text{Rem}(\text{pp}, C, i, m) := \text{VC.Update}(\text{pp}, C, i, -m)$ for the deterministic *removal* of value m from position i , and abbreviate $\text{VC.Verify}(\text{pp}, C, i, m, \pi)$ when $|S| = 1$.

Definition 2 (Position binding). VC is *position binding* if for every PPT adversary $\mathcal{A}$,

$$\text{Adv}_{\mathcal{A}, \text{VC}}^{\text{pos-bind}}(\lambda) := \Pr \left[\begin{array}{l} \text{pp} \leftarrow \text{VC.Setup}(1^\lambda, 1^n); \\ (C, i, m, \pi, m', \pi') \leftarrow \mathcal{A}(\text{pp}) : \\ m \neq m' \wedge \text{VC.Verify}(\text{pp}, C, i, m, \pi) = 1 \\ \wedge \text{VC.Verify}(\text{pp}, C, i, m', \pi') = 1 \end{array} \right] = \text{negl}(\lambda).$$

3.2 Collision-resistant hashing

Definition 3 (CRHF). A hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ is *collision resistant* if for every PPT adversary $\mathcal{A}$, $\text{Adv}_{\mathcal{A}, H}^{\text{cr}}(\lambda) := \Pr[(x, x') \leftarrow \mathcal{A}(1^\lambda) : x \neq x' \wedge H(x) = H(x')] = \text{negl}(\lambda)$.

3.3 Eviction policies

Definition 4 (Eviction policy). Let $\mathcal{S}$ be a totally ordered *score space* and $\mathcal{T}$ a *threshold space*. An *eviction policy* is an efficiently computable predicate $\text{Pol} : \mathcal{T} \times \mathcal{S} \rightarrow \{0, 1\}$ that is *antitone* in the score: $s \leq s' \Rightarrow \text{Pol}(\tau, s) \geq \text{Pol}(\tau, s')$ for all τ . The canonical *threshold policy* is $\text{Pol}_<(\tau, s) = 1 \iff s < \tau$, modelling “evict only low-attention positions”. We write $\text{Pol}_\tau(s)$ for $\text{Pol}(\tau, s)$.

Remark 1. The threshold policy captures per-position eligibility, the regime our construction proves secure. Set-relative policies (e.g. “evict the argmin” or “bottom- k ”) additionally require a verifiable minimality/sorting argument over the committed score vector; we treat these as an extension (Section 10).

4 Evictable Vector Commitments

An *evictable vector commitment* (EvictVC) commits jointly to a value vector and a score vector, supports in-place value updates, and supports *eviction*: the permanent, policy-checked removal of a position together with an append-only, tamper-evident record of what was removed.

Public state. A commitment is a tuple $C = (C_{\text{val}}, C_{\text{sc}}, \eta, t)$, where C_{val} commits to the *live* value vector, C_{sc} commits to the score vector, $\eta \in \{0, 1\}^\lambda$ is the *eviction-history digest*, and $t \in \mathbb{N}$ is the *epoch* (number of evictions so far). The evicted set $E \subseteq [n]$ (with live set $\Lambda = [n] \setminus E$) is public metadata, updated by verified evictions. Committer state st holds the vectors, Λ , and t .

History digest. For an ordered eviction log $\mathcal{L} = (r_1, \dots, r_\ell)$ with records $r_k = \langle i_k, v_k, k-1 \rangle$, define the hash chain

$$\eta_0 = H(\langle \text{genesis} \rangle), \quad \eta_k = H(\eta_{k-1} \parallel \langle r_k \rangle), \quad \text{HistDig}(\mathcal{L}) := \eta_\ell.$$

4.1 Syntax

Definition 5 (EvictVC). An evictable vector commitment for value space $\mathcal{V}$, score space $\mathcal{S}$ and policy Pol is a tuple of PPT algorithms $\Pi = (\text{Setup}, \text{Commit}, \text{Open}, \text{Verify}, \text{Update}, \text{Evict}, \text{VerifyEvict})$:

$\text{Setup}(1^\lambda, 1^n) \rightarrow \text{pp}$. Outputs public parameters for vectors of length up to n .

$\text{Commit}(\text{pp}, \mathbf{v}, \mathbf{s}) \rightarrow (C, \text{st})$. On $\mathbf{v} \in \mathcal{V}^n, \mathbf{s} \in \mathcal{S}^n$, outputs $C = (C_{\text{val}}, C_{\text{sc}}, \eta_0, 0)$ with $\Lambda = [n]$ and committer state st .

165 $\text{Open}(\text{pp}, \text{st}, S) \rightarrow \pi$. For $S \subseteq \Lambda$, outputs a proof for the live values $\{v_i\}_{i \in S}$.

166 $\text{Verify}(\text{pp}, C, S, \{v_i\}_{i \in S}, \pi) \rightarrow \{0, 1\}$. Verifies against C_{val} .

167 $\text{Update}(\text{pp}, \text{st}, i, \delta) \rightarrow (C', \text{st}')$. For $i \in \Lambda$, replaces v_i by $v_i + \delta$, sets $C'_{\text{val}} \leftarrow \text{VC.Update}(\text{pp}, C_{\text{val}}, i, \delta)$,
 168 leaving C_{sc}, η, t unchanged.

169 $\text{Evict}(\text{pp}, \text{st}, i, \tau) \rightarrow (C', \text{st}', \omega_i)$ or $\perp$. For $i \in \Lambda$: if $\text{Pol}_\tau(s_i) = 0$ output $\perp$; otherwise
 170 compute openings $\pi_i^{\text{val}}, \pi_i^{\text{sc}}$ of $C_{\text{val}}, C_{\text{sc}}$ at i ; set $C'_{\text{val}} \leftarrow \text{Rem}(\text{pp}, C_{\text{val}}, i, v_i)$, $C'_{\text{sc}} \leftarrow C_{\text{sc}}$,
 171 $\eta' \leftarrow \text{H}(\eta \parallel \langle i, v_i, t \rangle)$, $t' \leftarrow t + 1$, $\Lambda' \leftarrow \Lambda \setminus \{i\}$; output $C' = (C'_{\text{val}}, C'_{\text{sc}}, \eta', t')$, st' , and
 172 $\omega_i = (v_i, s_i, \pi_i^{\text{val}}, \pi_i^{\text{sc}})$.

173 $\text{VerifyEvict}(\text{pp}, C, i, \tau, \omega_i, C') \rightarrow \{0, 1\}$. Parse $\omega_i = (v, s, \pi^{\text{val}}, \pi^{\text{sc}})$. Output 1 iff: (a) $\text{VC.Verify}(\text{pp}, C_{\text{val}}, i, v, \pi^{\text{val}}) =$
 174 1; (b) $\text{VC.Verify}(\text{pp}, C_{\text{sc}}, i, s, \pi^{\text{sc}}) = 1$; (c) $\text{Pol}_\tau(s) = 1$; (d) $C'_{\text{val}} = \text{Rem}(\text{pp}, C_{\text{val}}, i, v)$ and
 175 $C'_{\text{sc}} = C_{\text{sc}}$; (e) $\eta' = \text{H}(\eta \parallel \langle i, v, t \rangle)$ and $t' = t + 1$.

176 *Remark 2* (Index consistency is structural). Checks (a)–(e) all refer to the *same* index i ,
 177 so an eviction of i cannot be justified by another position $j \neq i$'s score; this binding is
 178 syntactic and needs no computational assumption.

179 4.2 Correctness

180 **Definition 6** (Correctness). An execution is *admissible* if it is a **Commit** followed by any
 181 sequence of **Update**/**Evict** calls in which every call targets a currently live index and every
 182 **Evict** targets i with $\text{Pol}_\tau(s_i) = 1$ for the supplied τ . Π is *correct* if for all λ, n , all $(\mathbf{v}, \mathbf{s})$,
 183 and all admissible executions, with probability 1 over $\text{pp} \leftarrow \text{Setup}(1^\lambda, 1^n)$: (i) for every
 184 $S \subseteq \Lambda$, $\text{Verify}(\text{pp}, C, S, \{v_i\}_{i \in S}, \text{Open}(\text{pp}, \text{st}, S)) = 1$; and (ii) every $\text{Evict}(\text{pp}, \text{st}, i, \tau) \neq \perp$
 185 producing $(C', \text{st}', \omega_i)$ satisfies $\text{VerifyEvict}(\text{pp}, C, i, \tau, \omega_i, C') = 1$.

186 5 Security Model

187 We require three properties, each an experiment between a challenger and a PPT adversary
 188 $\mathcal{A}$; a parsed commitment is $C = (C_{\text{val}}, C_{\text{sc}}, \eta, t)$ and $\text{Adv}(\lambda) := \Pr[\text{Exp}(\lambda) = 1]$.

189 5.1 Eviction soundness

190 A verifying eviction may only remove a position whose committed score satisfies the policy.

Experiment $\text{Exp}_{\mathcal{A}, \Pi}^{\text{ev-snd}}(\lambda)$:

1. $\text{pp} \leftarrow \text{Setup}(1^\lambda, 1^n)$.
- 191 2. $(C, i, \tau, \omega_i, C', s^\dagger, \pi^\dagger) \leftarrow \mathcal{A}(\text{pp})$, where $\omega_i = (\hat{v}, \hat{s}, \hat{\pi}^{\text{val}}, \hat{\pi}^{\text{sc}})$.
3. Output 1 iff (a) $\text{VerifyEvict}(\text{pp}, C, i, \tau, \omega_i, C') = 1$ and (b)
 $\text{VC.Verify}(\text{pp}, C_{\text{sc}}, i, s^\dagger, \pi^\dagger) = 1$ with $\text{Pol}_\tau(s^\dagger) = 0$.

192 **Definition 7** (Eviction soundness). Π is *eviction sound* if $\text{Adv}_{\mathcal{A}, \Pi}^{\text{ev-snd}}(\lambda) = \text{negl}(\lambda)$ for every
 193 PPT $\mathcal{A}$.

194 5.2 Policy-binding

195 The value an eviction removes (and records) is the unique committed value at that position.

Experiment $\text{Exp}_{\mathcal{A}, \Pi}^{\text{pol-bind}}(\lambda)$:

1. $\text{pp} \leftarrow \text{Setup}(1^\lambda, 1^n)$.
2. $(C, i, \tau, \omega_i, C', v^\dagger, \pi^\dagger) \leftarrow \mathcal{A}(\text{pp})$, where $\omega_i = (\hat{v}, \hat{s}, \hat{\pi}^{\text{val}}, \hat{\pi}^{\text{sc}})$.
3. Output 1 iff (a) $\text{VerifyEvict}(\text{pp}, C, i, \tau, \omega_i, C') = 1$ and (b) $\text{VC.Verify}(\text{pp}, C_{\text{val}}, i, v^\dagger, \pi^\dagger) = 1$ with $v^\dagger \neq \hat{v}$.

Definition 8 (Policy-binding). Π is *policy-binding* if $\text{Adv}_{\mathcal{A}, \Pi}^{\text{pol-bind}}(\lambda) = \text{negl}(\lambda)$ for every PPT $\mathcal{A}$.

5.3 History tamper-evidence

The eviction history is an append-only log whose digest commits to the full ordered sequence of evictions.

Experiment $\text{Exp}_{\mathcal{A}, \Pi}^{\text{hist-te}}(\lambda)$:

1. $\text{pp} \leftarrow \text{Setup}(1^\lambda, 1^n)$.
2. $(\mathcal{L}, \mathcal{L}') \leftarrow \mathcal{A}(\text{pp})$, two eviction logs.
3. Output 1 iff $\mathcal{L} \neq \mathcal{L}'$ and $\text{HistDig}(\mathcal{L}) = \text{HistDig}(\mathcal{L}')$.

Definition 9 (History tamper-evidence). Π is *history tamper-evident* if $\text{Adv}_{\mathcal{A}, \Pi}^{\text{hist-te}}(\lambda) = \text{negl}(\lambda)$ for every PPT $\mathcal{A}$.

Definition 10 (Secure EvictVC). Π is *secure* if it is correct (Definition 6) and satisfies eviction soundness (Definition 7), policy-binding (Definition 8), and history tamper-evidence (Definition 9).

Remark 3 (What the three properties jointly guarantee). Together they certify each eviction end-to-end: the removed position was policy-eligible under the committed scores (eviction soundness), the removed value was its genuine committed value and the live state is faithfully shrunk (policy-binding), and the cumulative record of removals is immutable (history tamper-evidence). An auditor holding C , the public evicted set E , and the per-step evidence can verify the entire compression trajectory without trusting the prover.

6 The KZG Vector Commitment

We instantiate the position-binding vector commitment of Definition 1 with the Lagrange-basis KZG commitment [KZG10] over a multiplicative subgroup of roots of unity. Throughout this and the next two sections we index positions by $\{0, \dots, n-1\}$, identifying position i with the domain point ω^i ; n is a power of two.

6.1 Hardness assumption

Security rests on the q -Strong Diffie–Hellman assumption of Boneh and Boyen [BB04], stated in *structured* (SRS) form so that the reductions of Section 8 can perfectly simulate the public parameters, which publish $\mathbb{G}_2$ powers for subvector verification.

Assumption 6.1 (n -SDH, SRS form). For $\tau \xleftarrow{\$} \mathbb{F}_p^*$ and every PPT $\mathcal{B}$,

$$\text{Adv}_{\mathcal{B}}^{n\text{-SDH}}(\lambda) := \Pr \left[\begin{array}{l} (c, W) \leftarrow \mathcal{B}(\{\tau^k\}_1^n, \{\tau^k\}_2^n) : \\ c \in \mathbb{F}_p \setminus \{\tau\} \wedge W = \left[\frac{1}{\tau - c} \right]_1 \end{array} \right] = \text{negl}(\lambda).$$

6.2 Construction

Let $\omega \in \mathbb{F}_p$ be a primitive n -th root of unity, $\mathbb{D} = \{\omega^i\}_{i=0}^{n-1}$, and $\{\ell_i\}_{i=0}^{n-1}$ the Lagrange basis for $\mathbb{D}$ ($\ell_i(\omega^j) = \delta_{ij}$, $\deg \ell_i < n$).

Construction 6.1 (KZG vector commitment VC_{KZG}).

VC.Setup($1^\lambda, 1^n$). Sample $\tau \xleftarrow{\$} \mathbb{F}_p^*$ and output $\text{pp} = (\{\tau^k\}_1, \{\tau^k\}_2, \{L_i\}_{i=0}^{n-1}, \omega)$ with $L_i := [\ell_i(\tau)]_1$. The trapdoor τ is erased. The $\mathbb{G}_2$ powers support subvector verification; single-point verification uses only $[1]_2, [\tau]_2$.

VC.Commit($\text{pp}, \mathbf{m}$). Treat $\mathbf{m} = (m_0, \dots, m_{n-1})$ as evaluations on $\mathbb{D}$; the interpolant is $\phi(X) = \sum_i m_i \ell_i(X)$. Output $C = [\phi(\tau)]_1 = \sum_i m_i L_i$ and $\text{st} = \mathbf{m}$.

VC.Open(pp, st, S). Let $Z_S(X) = \prod_{i \in S} (X - \omega^i)$ and r_S the interpolant of $\{(\omega^i, m_i)\}_{i \in S}$ ($\deg < |S|$). Output $\pi_S = [\psi_S(\tau)]_1$ where $\psi_S = (\phi - r_S)/Z_S$. For $S = \{i\}$, $\psi_i = (\phi - m_i)/(X - \omega^i)$.

VC.Verify($\text{pp}, C, S, \{m_i\}_{i \in S}, \pi_S$). Recompute r_S, Z_S and accept iff $e(C - [r_S(\tau)]_1, [1]_2) = e(\pi_S, [Z_S(\tau)]_2)$. For $S = \{i\}$ this is $e(C - [m_i]_1, [1]_2) = e(\pi_i, [\tau]_2 - \omega^i [1]_2)$.

VC.Update(pp, C, i, δ). Output $C + \delta L_i$; and $\text{Rem}(\text{pp}, C, i, m) := C - m L_i$.

Both $[r_S(\tau)]_1$ and $[Z_S(\tau)]_2$ are computable by the verifier from the SRS, S , and the claimed values, so verification needs no secret state.

6.3 Correctness of openings

Lemma 1 (Opening identity). *For honestly produced C, π_S , $\text{VC.Verify}(\text{pp}, C, S, \{m_i\}_{i \in S}, \pi_S) = 1$.*

Proof. By interpolation $\phi(\omega^i) = m_i = r_S(\omega^i)$ for $i \in S$, so $Z_S \mid (\phi - r_S)$ and $\phi - r_S = \psi_S Z_S$. Evaluating at the trapdoor and using bilinearity, $e(C - [r_S(\tau)]_1, [1]_2) = [\phi(\tau) - r_S(\tau)]_T = [\psi_S(\tau) Z_S(\tau)]_T = e(\pi_S, [Z_S(\tau)]_2)$. $\square$

Lemma 2 (Homomorphic update). *$\text{VC.Update}(\text{pp}, C, i, \delta)$ is the commitment to $\mathbf{m}$ with m_i replaced by $m_i + \delta$, and $\text{Rem}(\text{pp}, C, i, m_i)$ is the commitment with m_i replaced by 0.*

Proof. $C + \delta L_i = [(\phi + \delta \ell_i)(\tau)]_1$, and $\phi + \delta \ell_i$ interpolates $\mathbf{m}$ with the i -th evaluation shifted by δ since $\ell_i(\omega^j) = \delta_{ij}$. Removal is $\delta = -m_i$. $\square$

Remark 4 (Position binding). VC_{KZG} is position binding under n -SDH; the reduction is Lemma 3.

7 The EvictVC Construction

We build Π_{KZG} by committing to the value and score vectors with two instances of VC_{KZG} under a shared pp , and binding the history with a hash chain. We write $\text{VC.Verify}_{\text{val}}$ and $\text{VC.Verify}_{\text{sc}}$ for verification against C_{val} and C_{sc} ; both are Construction 6.1's algorithm.

Construction 7.1 (Π_{KZG}). Fix a collision-resistant $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ and policy Pol .

Setup($1^\lambda, 1^n$). Output $\text{pp} \leftarrow \text{VC.Setup}(1^\lambda, 1^n)$.

Commit($\text{pp}, \mathbf{v}, \mathbf{s}$). Set $C_{\text{val}} \leftarrow \text{VC.Commit}(\text{pp}, \mathbf{v})$, $C_{\text{sc}} \leftarrow \text{VC.Commit}(\text{pp}, \mathbf{s})$, $\eta_0 \leftarrow H(\langle \text{genesis} \rangle)$, $t \leftarrow 0$, $\Lambda \leftarrow \{0, \dots, n-1\}$; output $C = (C_{\text{val}}, C_{\text{sc}}, \eta_0, 0)$, st .

Open/Verify. $\text{VC.Open} / \text{VC.Verify}_{\text{val}}$ on C_{val} .

263 **Update**($\text{pp}, \text{st}, i, \delta$). $C'_{\text{val}} \leftarrow \text{VC.Update}(\text{pp}, C_{\text{val}}, i, \delta)$, $v_i += \delta$.

264 **Evict**($\text{pp}, \text{st}, i, \tau_p$). If $\text{Pol}_{\tau_p}(s_i) = 0$ output $\perp$; else compute $\pi_i^{\text{val}} \leftarrow \text{VC.Open}(\text{pp}, \text{v}, \{i\})$,
 265 $\pi_i^{\text{sc}} \leftarrow \text{VC.Open}(\text{pp}, \text{s}, \{i\})$; set $C'_{\text{val}} \leftarrow C_{\text{val}} - v_i L_i$, $\eta' \leftarrow H(\eta \parallel \langle i, v_i, t \rangle)$, $t' \leftarrow t + 1$,
 266 $\Lambda' \leftarrow \Lambda \setminus \{i\}$; output $C' = (C'_{\text{val}}, C_{\text{sc}}, \eta', t')$, st' , $\omega_i = (v_i, s_i, \pi_i^{\text{val}}, \pi_i^{\text{sc}})$.

267 **VerifyEvict**($\text{pp}, C, i, \tau_p, \omega_i, C'$). Parse $\omega_i = (v, s, \pi^{\text{val}}, \pi^{\text{sc}})$; output 1 iff (a) $\text{VC.Verify}_{\text{val}}(\text{pp}, C_{\text{val}}, i, v, \pi^{\text{val}}) =$
 268 1; (b) $\text{VC.Verify}_{\text{sc}}(\text{pp}, C_{\text{sc}}, i, s, \pi^{\text{sc}}) = 1$; (c) $\text{Pol}_{\tau_p}(s) = 1$; (d) $C'_{\text{val}} = C_{\text{val}} - v L_i$ and $C'_{\text{sc}} = C_{\text{sc}}$;
 269 (e) $\eta' = H(\eta \parallel \langle i, v, t \rangle)$ and $t' = t + 1$.

270 The eviction threshold is written τ_p to distinguish it from the SRS trapdoor τ .

271 **Theorem 1** (Correctness). Π_{KZG} satisfies Definition 6.

272 *Proof.* (i) At every state $C_{\text{val}} = [\phi(\tau)]_1$ for the interpolant ϕ of the current value vector:
 273 **Commit** establishes this, **Update** preserves it and **Evict** zeroes position i , both by Lemma 2;
 274 hence any $S \subseteq \Lambda$ verifies by Lemma 1. (ii) If **Evict** $\neq \perp$ then $\text{Pol}_{\tau_p}(s_i) = 1$; checks (a),(b)
 275 hold by Lemma 1, (c) by eligibility, (d) since $C'_{\text{val}} = C_{\text{val}} - v_i L_i$ and C_{sc} is unchanged, and
 276 (e) since η', t' were produced by the same step **VerifyEvict** recomputes. $\square$

277 **Remark 5** (One-element batch audits). A verifier auditing any set S of live positions calls
 278 $\text{VC.Verify}_{\text{val}}$ on the single subvector proof $\pi_S = [\psi_S(\tau)]_1$: one $\mathbb{G}_1$ element, verified with
 279 two pairings, independent of $|S|$. This is the constant-size audit proof that distinguishes
 280 the scheme from Merkle multiproofs.

281 **Remark 6** (Amortized opening). Naively one opening costs $O(n \log n)$. By Feist–Khovratovich [FK23],
 282 all n single-point proofs are produced by a single $O(n \log n)$ group-FFT computation, i.e.
 283 $O(\log n)$ amortized, and are maintainable under updates in the same complexity.

284 The asymptotic costs of Π_{KZG} (n = vector length; $\mathbb{G}_1$ operations unless noted):

Operation	Time	Output / proof size
Setup (one-time)	$O(n)$ scalar mults + $O(n \log n)$ for $\{L_i\}$	pp of size $O(n)$
Commit	$O(n)$ (MSM)	one $\mathbb{G}_1$ element
286 Update / Evict (commit.)	$O(1)$	one $\mathbb{G}_1$ element
history step	$O(1)$ (one hash)	λ bits
Open (single / subvector)	$O(n \log n)$; $O(\log n)$ amort. (FK)	one $\mathbb{G}_1$ element
Verify / VerifyEvict	$O(1)$ (two pairings)	—

287 8 Security Proofs

288 The two binding-type properties reduce to position binding of VC_{KZG} under n -SDH; history
 289 tamper-evidence reduces to collision resistance. We write $\mathcal{B}$ for the reduction adversary.

290 8.1 Position binding of the KZG vector commitment

291 **Lemma 3** (Position binding from n -SDH). For every PPT $\mathcal{A}$ against position binding of
 292 VC_{KZG} (Definition 2) there is a PPT $\mathcal{B}$ with $\text{Adv}_{\mathcal{A}, \text{VC}_{\text{KZG}}}^{\text{pos-bind}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{n\text{-SDH}}(\lambda) + n/p$.

293 *Proof.* $\mathcal{B}$ receives an SRS-form n -SDH challenge $(\{\tau^k\}_1, \{\tau^k\}_2)_{k=0}^n$ and assembles
 294 pp as in Construction 6.1: it keeps the powers up to τ^{n-1} , fixes ω , and computes each
 295 $L_i = \sum_k c_{i,k} [\tau^k]_1$ from the public coefficients $\ell_i(X) = \sum_k c_{i,k} X^k$. This pp is distributed
 296 as in the real experiment, so $\mathcal{B}$ runs $\mathcal{A}(\text{pp}) \rightarrow (C, i, m, \pi, m', \pi')$.

Suppose $\mathcal{A}$ wins: $m \neq m'$ and, with $z := \omega^i$,

$$e(C - [m]_1, [1]_2) = e(\pi, [\tau]_2 - z[1]_2), \quad (1)$$

$$e(C - [m']_1, [1]_2) = e(\pi', [\tau]_2 - z[1]_2). \quad (2)$$

Dividing (1) by (2), the C -terms cancel and $e([m' - m]_1, [1]_2) = e(\pi - \pi', [\tau - z]_2)$. If $z = \tau$ the right side is the identity, forcing $m = m'$; hence $z \neq \tau$, an event whose complement we condition on (since $z \in \mathbb{D}$ and $\tau \xleftarrow{\$} \mathbb{F}_p^*$, $\tau \in \mathbb{D}$ has probability $\leq n/p$). With $\Delta := m' - m \neq 0$ and $W := \Delta^{-1}(\pi - \pi')$,

$$e(W, [\tau - z]_2) = e(\pi - \pi', [\tau - z]_2)^{\Delta^{-1}} = e([\Delta]_1, [1]_2)^{\Delta^{-1}} = [1]_T = e(g_1, g_2),$$

so by non-degeneracy $W = [1/(\tau - z)]_1$, and $\mathcal{B}$ outputs (z, W) . $\square$

8.2 Eviction soundness

Theorem 2 (Eviction soundness). *For every PPT $\mathcal{A}$ there is a PPT $\mathcal{B}$ with $\text{Adv}_{\mathcal{A}, \Pi_{\text{KZG}}}^{\text{ev-snd}}(\lambda) \leq \text{Adv}_{\mathcal{B}, \text{VC}_{\text{KZG}}}^{\text{pos-bind}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{n\text{-SDH}}(\lambda) + n/p$.*

Proof. $\mathcal{B}$ forwards pp to $\mathcal{A}$ and runs the eviction-soundness experiment, obtaining $(C, i, \tau_p, \omega_i, C', s^\dagger, \pi^\dagger)$ with $\omega_i = (\hat{v}, \hat{s}, \hat{\pi}^{\text{val}}, \hat{\pi}^{\text{sc}})$. If $\mathcal{A}$ wins, VerifyEvict accepts, so by checks (b),(c) $\text{VC.Verify}(\text{pp}, C_{\text{sc}}, i, \hat{s}, \hat{\pi}^{\text{sc}}) = 1$ with $\text{Pol}_{\tau_p}(\hat{s}) = 1$, while the winning condition gives $\text{VC.Verify}(\text{pp}, C_{\text{sc}}, i, s^\dagger, \pi^\dagger) = 1$ with $\text{Pol}_{\tau_p}(s^\dagger) = 0$. As $\text{Pol}_{\tau_p}(\hat{s}) \neq \text{Pol}_{\tau_p}(s^\dagger)$, $\hat{s} \neq s^\dagger$, so $(C_{\text{sc}}, i, \hat{s}, \hat{\pi}^{\text{sc}}, s^\dagger, \pi^\dagger)$ breaks position binding on C_{sc} . Apply Lemma 3. $\square$

8.3 Policy-binding

Theorem 3 (Policy-binding). *For every PPT $\mathcal{A}$ there is a PPT $\mathcal{B}$ with $\text{Adv}_{\mathcal{A}, \Pi_{\text{KZG}}}^{\text{pol-bind}}(\lambda) \leq \text{Adv}_{\mathcal{B}, \text{VC}_{\text{KZG}}}^{\text{pos-bind}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{n\text{-SDH}}(\lambda) + n/p$.*

Proof. $\mathcal{B}$ runs $\mathcal{A}$ and obtains $(C, i, \tau_p, \omega_i, C', v^\dagger, \pi^\dagger)$ with $\omega_i = (\hat{v}, \hat{s}, \hat{\pi}^{\text{val}}, \hat{\pi}^{\text{sc}})$. If $\mathcal{A}$ wins, VerifyEvict accepts, so by check (a) $\text{VC.Verify}(\text{pp}, C_{\text{val}}, i, \hat{v}, \hat{\pi}^{\text{val}}) = 1$, while the winning condition gives $\text{VC.Verify}(\text{pp}, C_{\text{val}}, i, v^\dagger, \pi^\dagger) = 1$ with $v^\dagger \neq \hat{v}$. Thus $(C_{\text{val}}, i, \hat{v}, \hat{\pi}^{\text{val}}, v^\dagger, \pi^\dagger)$ breaks position binding on C_{val} . Apply Lemma 3. $\square$

Remark 7. Checks (a) and (d) together pin down the post-eviction state: $\hat{v}$ is the unique committed value at i and $C'_{\text{val}} = C_{\text{val}} - \hat{v}L_i$, so position i is provably zeroed and cannot later be opened to a nonzero value. Eviction is a genuine, irreversible deletion, not a relabelling.

8.4 History tamper-evidence

Theorem 4 (History tamper-evidence). *If H is collision resistant then for every PPT $\mathcal{A}$ there is a PPT $\mathcal{B}$ with $\text{Adv}_{\mathcal{A}, \Pi_{\text{KZG}}}^{\text{hist-te}}(\lambda) \leq \text{Adv}_{\mathcal{B}, H}^{\text{cr}}(\lambda)$.*

Proof. $\mathcal{B}$ runs $\mathcal{A}$ to get $\mathcal{L} = (r_1, \dots, r_\ell)$, $\mathcal{L}' = (r'_1, \dots, r'_{\ell'})$ with $\mathcal{L} \neq \mathcal{L}'$ and equal digest, and walks the two chains backward maintaining “current digests equal”. At a step comparing $\eta_a = \eta'_b$: if $a, b \geq 1$ then $H(\eta_{a-1} \parallel \langle r_a \rangle) = H(\eta'_{b-1} \parallel \langle r'_b \rangle)$; if the pre-images differ, output them as a collision, else (fixed-length chaining value and prefix-free $\langle \cdot \rangle$) $\eta_{a-1} = \eta'_{b-1}$, $r_a = r'_b$, and recurse. If exactly one of a, b is 0, the two pre-images differ by domain separation of the genesis tag, yielding a collision. If $a = b = 0$, every record matched, so $\mathcal{L} = \mathcal{L}'$, contradicting $\mathcal{L} \neq \mathcal{L}'$. Hence a collision is always emitted. $\square$

8.5 Main theorem

Theorem 5 (Security of Π_{KZG}). *If n -SDH holds and H is collision resistant, then Π_{KZG} is a secure evictable vector commitment (Definition 10), with $\text{Adv}^{\text{ev-snd}}, \text{Adv}^{\text{pol-bind}} \leq \text{Adv}_{\mathcal{B}}^{n\text{-SDH}} + n/p$ and $\text{Adv}^{\text{hist-te}} \leq \text{Adv}_{\mathcal{B}, H}^{\text{cr}}$.*

Proof. Correctness is Theorem 1; the three properties are Theorems 2, 3, 4, negligible under n -SDH (via Lemma 3, with n/p negligible for λ -bit p) and collision resistance respectively. $\square$

Remark 8 (Tightness and instantiation). All reductions invoke their adversary once and preserve advantage up to the additive n/p from the trapdoor-collision event, $\leq 2^{-\lambda + \log n}$ for $p \approx 2^\lambda$. Over BLS12-381 (255-bit p , 256-bit H) the bounds are below 2^{-128} for all $n \leq 2^{40}$.

9 Evaluation

Implementation. We implement Π_{KZG} twice: a Python reference (`py_ecc`, BLS12-381) covering the full eviction protocol—policy-checked eviction, tamper-evident hash-chain history, and single-element subvector audits—and a performance backend in Rust on `arkworks` (`ark-bls12-381`). The Rust backend realizes commitment, $O(1)$ update/eviction against precomputed Lagrange-basis commitments, single-point opening and pairing verification, and the Feist–Khovratovich all-proofs routine.

Setup. Measurements are taken single-threaded on an Apple M-series laptop with a Rust release build, for vector (cache) lengths n from 2^{10} to 2^{16} .

9.1 Commitment maintenance

Table 1 reports per-operation cost as the cache length n ranges over a $64\times$ window. The defining property of the scheme is visible directly in the *update* column: eviction and value update cost $127\text{--}142\ \mu\text{s}$ with no upward trend, the variation being measurement noise rather than growth in n . This is forced by the construction. An update is $C'_{\text{val}} = C_{\text{val}} \pm \delta L_i$: one scalar multiplication and one point addition against the precomputed Lagrange-basis commitment L_i , touching a single group element and never reading or recombining the other $n - 1$ positions. The cost is therefore independent of n by construction, and an eviction—a removal $C_{\text{val}} - v L_i$ plus one hash for the history step—inherits the same $O(1)$ profile. For comparison, recommitting from scratch is $O(n)$ and even an incremental Merkle update is $O(\log n)$; only the homomorphic group update is genuinely constant-time, which is what makes per-token eviction viable in a streaming setting where the cache is recompressed continuously.

Verification is likewise flat (2.57–2.67 ms): it is two pairings whose cost is fixed by the curve, not by the cache, so an audit of any single position—or, by Remark 5, of any *set* of positions—verifies in constant time. Commitment and opening grow roughly linearly (about $3.3\text{--}3.4\times$ per $4\times$ increase in n), reflecting the multi-scalar multiplication that dominates the $O(n \log n)$ interpolation transform at these sizes. **Setup** is a one-time $O(n)$ cost; in deployment it is a powers-of-tau ceremony artifact loaded once and never appears on the per-operation path. The measurements are single-threaded; what transfers across hardware is the *shape*—flat update, constant verification, near-linear commit/open—not the absolute constants.

Table 1: Per-operation cost of Π_{KZG} (BLS12-381, single thread). Eviction/update is $O(1)$ and verification is constant; commit/open are $O(n \log n)$ and setup is a one-time $O(n)$ cost.

n	Setup (ms)	Commit (ms)	update (μs)	Open (ms)	Verify (ms)
2^{10}	167.3	18.2	141.5	18.3	2.57
2^{12}	622.4	59.2	126.8	59.2	2.57
2^{14}	2487.6	200.8	135.9	200.5	2.60
2^{16}	9954.7	681.0	135.9	683.8	2.67

9.2 Audit proof size

Figure 1 plots batch-audit proof size against the number of audited positions k at a fixed cache of $n = 2^{14}$. The EvictVC curve is a horizontal line at 48 bytes: a batch audit of any set S is the single subvector opening $\pi_S = [\psi_S(\tau)]_1$, and $\psi_S = (\phi - r_S)/Z_S$ is one polynomial whose commitment is one $\mathbb{G}_1$ element—a compressed BLS12-381 point—no matter how many positions S contains or how large n is. The Merkle baseline, by contrast, carries one authentication path of $\lceil \log_2 n \rceil$ sibling hashes (32 bytes each) per audited leaf: even a single-position proof is 448 bytes at $n = 2^{14}$ (14 nodes), already $9\times$ the KZG proof, and a k -position multiproof grows with k until it reaches hundreds of kilobytes.

The Merkle curve rises and then falls as $k \rightarrow n$: when nearly all leaves are audited the paths share most internal nodes, so the number of *distinct* authentication nodes shrinks (auditing all n leaves needs none—the auditor holds the whole tree). This is a correct artifact of multiproof deduplication; the operative comparison is the realistic regime $k \ll n$, where a verifier spot-checks a sample of cached positions. There Merkle is essentially linear in k while EvictVC is constant, a one- to three-order-of-magnitude reduction over the plotted range. Aggregation also collapses verification: the k -position audit is still two pairings, against the Merkle verifier’s $O(k \log n)$ hashing. We stress that the advantage is in proof size and aggregation, not single-update latency—a Merkle leaf update is a handful of hashes and is in fact cheaper than one group operation; EvictVC’s case rests on constant-size, aggregatable audits and the $O(1)$ commitment maintenance of Table 1, not on beating a hash with a pairing.

9.3 Amortized proof generation

Table 2 contrasts the Feist–Khovratovich all-proofs routine with the naive baseline of computing one opening per position. The two scale differently in the most consequential way. Naively, a single opening costs $O(n \log n)$, so producing a proof for every position costs $O(n^2 \log n)$; the “naive all” column is simply n times the measured single-open cost. Feist–Khovratovich instead obtains all n proofs from one $O(n \log n)$ group-FFT pass, an $O(\log n)$ amortized cost per position. This shows up in the table as a qualitative difference in the per-proof trend: FK’s amortized cost creeps only from 3.1 to 4.7 ms across 2^8 to 2^{12} (a logarithmic drift), while the naive single-open cost climbs roughly linearly (6 to 59 ms over the same window). The crossover is immediate—at every measured size FK’s amortized per-proof already undercuts a single naive opening (at 2^{12} , 4.7 ms versus 59 ms, a $12\times$ gap)—and the ratio widens with n .

At full-cache scale the separation becomes decisive. A single opening at $n = 2^{16}$ costs ≈ 684 ms (Table 1), so producing all 2^{16} proofs naively would take $684 \text{ ms} \times 2^{16} \approx 4.5 \times 10^4 \text{ s}$, about 12.5 hours; the FK routine, extrapolating its $O(n \log n)$ trend, completes the same task in roughly 7 minutes single-threaded, and its FFT parallelizes cleanly across cores. Beyond a handful of openings, FK is therefore not merely faster but the only feasible way to serve whole-cache audits. One honest caveat: FK’s absolute per-proof cost is milliseconds,

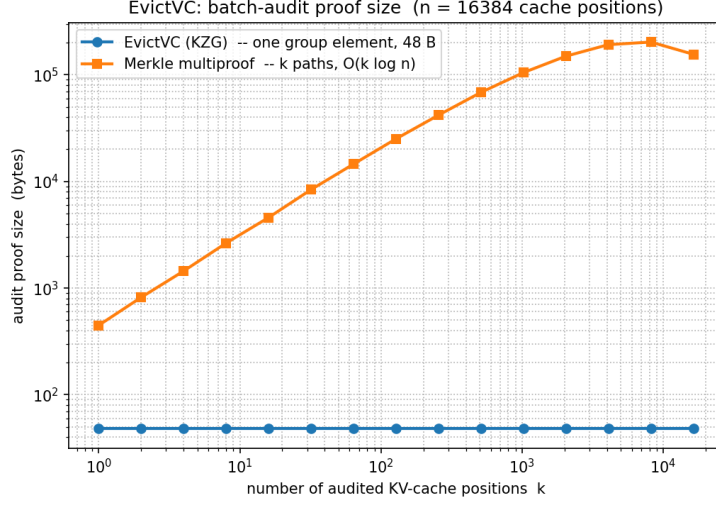


Figure 1: Batch-audit proof size versus number of audited positions k ($n = 2^{14}$). EvictVC’s proof is a constant 48-byte group element; the Merkle multiproof grows with k .

Table 2: Feist–Khovratovich all-proofs versus naive per-position opening (BLS12-381, single thread). FK’s amortized per-proof cost grows only logarithmically; the naive all-proofs estimate is $n \times$ a single opening.

n	FK all (ms)	FK / proof (μ s)	naive open (ms)	naive all (est., s)
2^8	800.7	3127.8	6.13	1.6
2^{10}	3980.3	3887.0	18.32	18.8
2^{12}	19095.5	4662.0	59.21	242.5

not microseconds, because the group-FFT performs $O(n \log n)$ scalar multiplications by roots of unity; the win is the asymptotic $O(n \log n)$ -versus- $O(n^2)$ separation in total work—realized here as hours versus minutes—rather than a small per-proof constant.

Summary. The three measurements validate the design end-to-end and along independent axes. Constant-time eviction (Table 1) makes it practical to apply the commitment per evicted token in a streaming compressor, where the cache is rewritten continuously. Constant-size, aggregatable audits (Figure 1) make verification scale *in* the number of checked positions rather than against it, so an auditor can sample as aggressively as it likes for a fixed 48-byte proof and two pairings. And amortized proof generation (Table 2) makes serving openings for the entire, possibly-evicted cache feasible at a scale where the naive approach is hours out of reach. None of these rests on a faster primitive than the alternatives in isolation; each follows from the algebraic structure—homomorphic $O(1)$ updates, one-element subvector openings, and batched proof computation—that the KZG instantiation provides and a hash-based commitment does not.

10 Conclusion

We introduced evictable vector commitments, a primitive that brings permanent, policy-checked deletion—with a tamper-evident history—to the vector commitments underlying verifiable LLM inference, and gave a KZG construction with $O(1)$ eviction, single-element

batch-audit proofs, and quasi-linear all-proof maintenance, proven secure under n -SDH and collision resistance.

Several extensions are natural. *Set-relative policies* such as $\arg \min$ /bottom- k eviction require a verifiable minimality or sorting argument over the committed score vector, layered on the threshold policy proven here. *Succinct history membership* would let an auditor prove a single past eviction without replaying the whole log, e.g. by replacing the hash chain with a Merkle Mountain Range. *Confidential eviction* could hide the evicted values—revealing only their commitments and update proofs—when the dropped state is sensitive. Finally, integrating EvictVC into an end-to-end verifiable-inference pipeline over a real streaming KV-cache compressor is the natural systems follow-up.

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